

Target validation paths — gist-sdh-multifocal-resected-m1-k4n8

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

The case hinges on whether the SDH-deficient GIST phenotype actually holds. Biallelic SDHA inactivation at ~85% combined VAF should destabilize the SDH complex and abolish SDHB staining, and SDHB IHC reads as intact. That paradox is the gate. Resolving it via a paired SDHB re-cut with positive internal control, an orthogonal SDHA IHC, and a germline SDHA panel locks in (or refutes) the driver call that controls enrollment on belzutifan via NCT04924075 and on the temozolomide trials NCT03556384 and NCT05661643. If both the IHC arbitration and the germline workup come back unsupportive, the trial-route options foreclose and the conversation about KIT/PDGFRA-WT systemic care for first recurrence belongs with the treating team rather than this report.

SDHA

Five workup steps stack on one tissue pull. Essential: SDHB IHC re-cut (clone 21A11 or BSB-131) with an explicit positive internal control vessel in the field, reviewed by a soft-tissue or GI pathologist who routinely sees SDH-deficient tumors; paired SDHA IHC (clone 2E3 or D6J9M) on the same block, which is specifically lost in SDHA-driven cases and arbitrates the SDHB-intact paradox more directly than re-staining SDHB alone; germline SDHA full-gene sequencing with del/dup analysis as part of a hereditary paraganglioma-pheochromocytoma panel covering all 10 PGL/PCC genes, which resolves whether SDHA L306P (VAF 44%) is germline or fully somatic and triggers cascade testing of first-degree relatives if germline; baseline paraganglioma surveillance with plasma free metanephrines by LC-MS/MS plus rapid-sequence whole-body MRI from skull base to pelvis, which is the patient-safety floor for any SDHx carrier going to anesthesia for any reason; and tumor tissue NGS on a comprehensive FDA-cleared panel (FoundationOne CDx or equivalent) to confirm or refute the ctDNA-only SDHA E350fs, PIK3CA R93W, and MAP2K1 P124S calls. High-priority: Signatera tumor-informed ctDNA MRD using an FFPE-derived bespoke panel as a recurrence-surveillance adjunct to imaging (informative, not therapy-triggering). Medium-priority: SDHC promoter methylation analysis as the fallback driver-call assay if the SDHA workup collapses, ordered through the NIH Pediatric and wild-type GIST Clinic since clinically-available testing is limited to academic referral labs; MGMT promoter methylation on archival FFPE, deferred until recurrence is anticipated or temozolomide enters the active discussion. Bundling the IHC re-cut, SDHA IHC, tumor NGS, and Signatera baseline build on a single archival block keeps the operational sequence to one tissue pull.

PIK3CA R93W

Tissue NGS confirmation on a comprehensive FDA-cleared panel (FoundationOne CDx, Tempus xT CDx, or equivalent) of the ctDNA-only call. R93W sits outside the validated alpelisib companion-diagnostic hotspots (E542K, E545K/A/D/G, H1047R/Y/L, C420R) and the variant has not been corroborated on tumor tissue, so any PI3K α -inhibitor discussion or trial enrollment is gated on this step. Same comprehensive panel resolves the MAP2K1 and SDHA E350fs questions on one requisition.

MAP2K1 P124S

Same tissue NGS panel as PIK3CA. P124 is a recognized MEK1 activating site (P124L/Q/S) with documented but heterogeneous MEK-inhibitor response, but the ctDNA-only call cannot anchor MEK-inhibitor consideration until clonality in tumor is established. Without that, MEK monotherapy in MAP2K1-mutant disease is a stretch outside histiocytosis and melanoma even at the best of times.

POLE R1679C

Cancer-predisposition genetic counseling referral with R1679C variant reclassification review. R1679 sits outside the canonical POLE proofreading hotspots (P286R, V411L, S459F) and the tumor is TMB-low and MSS, so the polymerase-proofreading-associated polyposis phenotype is not supported. Counseling resolves the variant on the right side of the ledger (most likely VUS, no current ICI rationale, no polyposis surveillance trigger) and folds the SDHA germline result into the same family-risk conversation rather than scheduling two staggered ones.

Where to order these assays

Assay	Provider	Decision gated	Contact
Germline SDHA full-gene sequencing + del/dup (hereditary PGL/PCC panel)	Labcorp Genetics (preferred) (Invitae Hereditary Panel)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384 / NCT05661643); triggers cascade testing of first-degree relatives.	test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037
Germline SDHA full-gene sequencing + del/dup (hereditary PGL/PCC panel)	GeneDx (GeneDx Panel)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384 / NCT05661643); triggers cascade testing of first-degree relatives.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline SDHA full-gene sequencing + del/dup (hereditary PGL/PCC panel)	Ambry Genetics (PGLNext)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384 / NCT05661643); triggers cascade testing of first-degree relatives.	test info · 15 Argonaut, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline SDHA full-gene sequencing + del/dup (hereditary PGL/PCC panel)	Mayo Clinic Laboratories (PGLPN / Hereditary Gene Panel)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384 / NCT05661643); triggers cascade testing of first-degree relatives.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Tumor tissue NGS confirmation of PIK3CA R93W	Foundation Medicine (preferred) (FoundationOne CDx)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor tissue NGS confirmation of PIK3CA R93W	Tempus Labs (Tempus xT CDx)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor tissue NGS confirmation of PIK3CA R93W	Caris Life Sciences (MI Cancer Seek / MI Tumor Seek Hybrid)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Tumor tissue NGS confirmation of PIK3CA R93W	NeoGenomics Laboratories (NeoTYPE Comprehensive Tumor Profile)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907

Tumor tissue NGS confirmation of MAP2K1 P124S	Foundation Medicine (preferred) (FoundationOne CDx)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor tissue NGS confirmation of MAP2K1 P124S	Tempus Labs (Tempus xT CDx)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor tissue NGS confirmation of MAP2K1 P124S	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tumor tissue NGS confirmation of MAP2K1 P124S	Caris Life Sciences (MI Cancer Seek)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
SDHB IHC re-cut with positive internal control	Mayo Clinic Laboratories (preferred) (SDHB Immunostain test code SDHB / 70550, technical component only)	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
SDHB IHC re-cut with positive internal control	Memorial Sloan Kettering Pathology Consultation Service	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
SDHB IHC re-cut with positive internal control	Brigham and Women's Hospital Department of Pathology (Dana-Farber / BWH GIST referral)	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 75 Francis Street, Boston, MA 02115 · 1-617-732-7510
SDHB IHC re-cut with positive internal control	NeoGenomics Laboratories (SDHB IHC)	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
SDHA IHC (clone 2E3 or D6J9M)	Mayo Clinic Laboratories (preferred) (SDHA Immunostain)	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
SDHA IHC (clone 2E3 or D6J9M)	Memorial Sloan Kettering Pathology Consultation Service	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
SDHA IHC (clone 2E3 or D6J9M)	Brigham and Women's Hospital Department of Pathology	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 75 Francis Street, Boston, MA 02115 · 1-617-732-7510

SDHA IHC (clone 2E3 or D6J9M)	NeoGenomics Laboratories (SDHA IHC)	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Baseline PGL / PCC surveillance: whole-body MRI + Ga-68 DOTATATE PET/CT	Treating-center academic radiology (preferred) (NCI-designated center with a NET program)	Paraganglioma / pheochromocytoma surveillance frequency and modality choice for an SDHx-deficient patient.	varies by region
Baseline PGL / PCC surveillance: whole-body MRI + Ga-68 DOTATATE PET/CT	RadNet (national imaging network)	Paraganglioma / pheochromocytoma surveillance frequency and modality choice for an SDHx-deficient patient.	test info · 1510 Cotner Avenue, Los Angeles, CA 90025 · 1-310-478-7808
Baseline PGL / PCC surveillance: whole-body MRI + Ga-68 DOTATATE PET/CT	NIH Clinical Center (Pacak laboratory)	Paraganglioma / pheochromocytoma surveillance frequency and modality choice for an SDHx-deficient patient.	test info · 10 Center Drive, Bethesda, MD 20892 · 1-800-411-1222
Plasma free metanephrines + normetanephrines (LC-MS/MS) and / or 24-hour urine fractionated metanephrines	Mayo Clinic Laboratories (preferred) (PMETF / Metanephrines, Fractionated, Free, Plasma)	Functional paraganglioma / pheochromocytoma screen; informs anesthesia-safety planning and surveillance interval.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Plasma free metanephrines + normetanephrines (LC-MS/MS) and / or 24-hour urine fractionated metanephrines	Quest Diagnostics (Metanephrines, Free, Plasma (LC-MS/MS))	Functional paraganglioma / pheochromocytoma screen; informs anesthesia-safety planning and surveillance interval.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Plasma free metanephrines + normetanephrines (LC-MS/MS) and / or 24-hour urine fractionated metanephrines	Labcorp (Metanephrines, Fractionated, Plasma Free)	Functional paraganglioma / pheochromocytoma screen; informs anesthesia-safety planning and surveillance interval.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Plasma free metanephrines + normetanephrines (LC-MS/MS) and / or 24-hour urine fractionated metanephrines	ARUP Laboratories (Metanephrines, Fractionated, Free, Plasma by LC-MS/MS)	Functional paraganglioma / pheochromocytoma screen; informs anesthesia-safety planning and surveillance interval.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
Cancer-predisposition genetic counseling referral (POLE R1679C reclassification + SDHA cascade testing)	NSGC Find-a-Genetic-Counselor Directory (preferred)	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 330 N Wabash Ave, Suite 2000, Chicago, IL 60611 · 1-312-321-6834
Cancer-predisposition genetic counseling referral (POLE R1679C reclassification + SDHA cascade testing)	Dana-Farber Cancer Genetics and Prevention Program	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 450 Brookline Avenue, Boston, MA 02215 · 1-617-632-2178
Cancer-predisposition genetic counseling referral (POLE R1679C reclassification + SDHA cascade testing)	MD Anderson Clinical Cancer Genetics Program	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789

Cancer-predisposition genetic counseling referral (POLE R1679C reclassification + SDHA cascade testing)	Memorial Sloan Kettering Clinical Genetics Service	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 1275 York Avenue, New York, NY 10065 · 1-646-888-4050
Tumor tissue NGS for SDHA E350fs (same comprehensive panel as PIK3CA / MAP2K1)	Foundation Medicine (preferred) (FoundationOne CDx)	Clonal-architecture mapping; no direct intervention gating since biallelic SDHA inactivation already established.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor tissue NGS for SDHA E350fs (same comprehensive panel as PIK3CA / MAP2K1)	Tempus Labs (Tempus xT CDx)	Clonal-architecture mapping; no direct intervention gating since biallelic SDHA inactivation already established.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor-informed ctDNA MRD (Signatera) or tumor-naive comprehensive ctDNA panel	Natera (preferred) (Signatera tumor-informed ctDNA MRD)	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090
Tumor-informed ctDNA MRD (Signatera) or tumor-naive comprehensive ctDNA panel	Guardant Health (Guardant360 (tumor-naive comprehensive ctDNA))	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
Tumor-informed ctDNA MRD (Signatera) or tumor-naive comprehensive ctDNA panel	Foundation Medicine (FoundationOne Liquid CDx)	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor-informed ctDNA MRD (Signatera) or tumor-naive comprehensive ctDNA panel	Tempus Labs (Tempus xF / xF+)	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
SDHC promoter methylation analysis (pyrosequencing or methylation-specific PCR)	NIH Pediatric and Wild-Type GIST Clinic (preferred)	dSDH-GIST sub-driver call if SDHA inactivation is not confirmed; does not gate therapy by itself.	test info · 10 Center Drive, Bethesda, MD 20892 · 1-240-760-6000
SDHC promoter methylation analysis (pyrosequencing or methylation-specific PCR)	University Hospital Mannheim (Wegert / Hofmann group)	dSDH-GIST sub-driver call if SDHA inactivation is not confirmed; does not gate therapy by itself.	Theodor-Kutzer-Ufer 1-3, 68167 Mannheim, Germany
MGMT promoter methylation (pyrosequencing or methylation-specific PCR)	Mayo Clinic Laboratories (preferred) (MGMTM / MGMT Promoter Methylation, Tumor)	Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
MGMT promoter methylation (pyrosequencing or methylation-specific PCR)	Quest Diagnostics (MGMT Methylation Analysis)	Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MGMT promoter methylation (pyrosequencing or methylation-specific PCR)	Labcorp (MGMT Promoter Methylation)	Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

MGMT promoter methylation
(pyrosequencing or
methylation-specific PCR)

NeoGenomics Laboratories
(MGMT Methylation by PCR)

Temozolomide-response
prediction at recurrence
(NCT03556384, NCT05661643).

test info · 9490 NeoGenomics
Way, Fort Myers, FL 33912 ·
1-866-776-5907

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
Germline SDHA full-gene sequencing + del/dup (hereditary panel)	Resolves whether SDHA L306P (VAF 44%) is germline (single-hit carrier with a second somatic event) or fully somatic. Locks in the SDH-deficient driver call gating the temozolomide and belzutifan trials and triggers first-degree-relative cascade testing per the 2025 Florou review.	Labcorp Genetics (Invitae Hereditary Panel) · test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037	5-10 mL whole blood (EDTA) or saliva kit
Tumor tissue NGS confirmation of PIK3CA R93W (FoundationOne CDx, Tempus xT, or equivalent)	R93W sits outside the validated alpelisib companion-diagnostic hotspots and the ctDNA-only call has not been corroborated on tumor tissue. Confirms whether the variant is a real subclonal alteration or a ctDNA artifact and frames any PI3K α -inhibitor discussion.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable
Tumor tissue NGS confirmation of MAP2K1 P124S (comprehensive panel)	P124 is a recognized MEK1 activating site with documented but heterogeneous MEK-inhibitor response. The variant is currently ctDNA-only; tissue confirmation establishes whether it is clonal in the resected tumor or low-allele-fraction subclonal / artifactual.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable

SDHB IHC re-cut with positive internal control (clone 21A11 or BSB-131)	Biallelic SDHA inactivation at ~85% combined VAF should destabilize the SDH complex and abolish SDHB staining; retained SDHB in this case conflicts with the molecular call. A re-cut with explicit positive internal control resolves the most likely technical false-negative before the dSDH-GIST diagnosis anchors trial enrollment.	Mayo Clinic Laboratories (SDHB Immunostain test code SDHB / 70550) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	1 unstained slide from archival FFPE (block returns to pathology)
SDHA IHC (clone 2E3 or D6J9M) on archival FFPE	SDHA IHC is specifically lost in tumors with biallelic SDHA inactivation while retained in SDHB/C/D or SDHC-methylation-driven cases. Arbitrates the SDHB-intact / SDHA-biallelic conflict more directly than re-staining SDHB alone.	Mayo Clinic Laboratories (SDHA Immunostain) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	1 unstained slide from archival FFPE
Baseline PGL / PCC surveillance: rapid-sequence whole-body MRI plus Ga-68 DOTATATE PET/CT	SDHx carriers carry a lifetime paraganglioma / pheochromocytoma risk warranting imaging surveillance even when the index tumor is a GIST. Current Endocrine Society 2014 / ENS@T consensus recommends biennial whole-body MRI from skull base to pelvis with Ga-68 DOTATATE as the most sensitive functional tracer. Clean baseline now also serves as the recurrence-monitoring comparator.	Treating-center academic radiology at an NCI-designated center with an established NET / PGL program · varies by region	no specimen; imaging only
Plasma free metanephrines + normetanephrines (LC-MS/MS) plus 24-hour urine fractionated metanephrines	An undetected functional paraganglioma in an SDHx carrier going to surgery or anesthesia is a hypertensive-crisis hazard. Biochemical screening pairs with imaging as the standard SDHx workup per the Endocrine Society 2014 guideline; plasma free metanephrines by LC-MS/MS is the most sensitive single test.	Mayo Clinic Laboratories (PMETF / Metanephrines, Fractionated, Free, Plasma) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	10 mL plasma (EDTA) and / or 24-hour urine collection in acid-preserved container

genetic counseling referral with POLE R1679C reclassification review	R1679 sits outside the canonical POLE proofreading hotspots and the tumor is TMB-low / MSS, so the polyposis phenotype is not supported. Counseling resolves the variant on the right side of the ledger and folds the SDHA germline result into one coherent family-risk discussion.	NSGC Find-a-Genetic-Counselor Directory · test info · 330 N Wabash Ave, Suite 2000, Chicago, IL 60611 · 1-312-321-6834	no specimen required; counselor reviews the existing germline report
Tumor tissue NGS for SDHA E350fs (same comprehensive panel as PIK3CA / MAP2K1)	E350fs was called from ctDNA but not seen in the tumor NGS that already reported L306P and Y408N. Confirming or refuting clarifies whether this is a true third SDHA hit, a subclonal event in unsampled tissue, or a ctDNA artifact. Incremental rather than gating since biallelic inactivation is already established.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with PIK3CA / MAP2K1 block
Tumor-informed ctDNA MRD assay (Signatera) or tumor-naïve comprehensive ctDNA panel	Post-R0 with no measurable disease and the existing ctDNA call surface already informative (SDHA E350fs, PIK3CA R93W, MAP2K1 P124S). Serial ctDNA gives a lead-time signal for recurrence ahead of imaging. GIST has weaker prospective MRD validation than colorectal or lung; frame as informative rather than gating.	Natera (Signatera tumor-informed ctDNA MRD) · test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090	5-10 mL plasma (Streck tube); Signatera baseline also needs an FFPE block

SDHC promoter methylation analysis (pyrosequencing or PCR)	Fallback driver-call assay if the SDHA story collapses on the SDHA IHC / germline / E350fs workup. SDHC promoter epimutation is the next-most-likely Carney-triad-pattern driver and most dSDH-GIST trials accept methylation-driven cases as SDH-deficient. Clinically-available testing is limited to academic referral labs.	NIH Pediatric and Wild-Type GIST Clinic · test info · 10 Center Drive, Bethesda, MD 20892 · 1-240-760-6000	FFPE block or 10 unstained slides
MGMT promoter methylation (pyrosequencing or PCR)	Recent data show preferential MGMT promoter hypermethylation in SDH-deficient wild-type GIST and a corresponding rationale for temozolomide activity. If the temozolomide trials become the leading systemic option at recurrence, MGMT status helps predict response. Does not gate enrollment but adds depth once disease becomes measurable.	Mayo Clinic Laboratories (MGMTM / MGMT Promoter Methylation, Tumor) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	FFPE block or 10 unstained slides; bundle with SDHC methylation order if both are needed

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.